

Information Systems

Data Base Management System

IDENTIFICATION

CODE : IF-3-BDR
ECTS : 3.0

HOURS

Lectures : 12.0 h
Seminars : 10.0 h
Laboratory : 20.0 h
Project : 0.0 h
Teacher-student
contact : 42.0 h
Personal work : 35.0 h
Total : 77.0 h

ASSESSMENT METHOD

Labor course evaluation [multiple-choice questionnaire, report],
Final exam (1:30)

TEACHING AIDS

Lecture slides

TEACHING LANGUAGE

French

CONTACT

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AIMS

With an approach focused on the development of applications using databases, the module addresses the data management issues and the principles of Relational Database Management Systems, their motives and what they can bring in the global landscape of information systems.

It contributes to the acquisition and development of specific skills defined in the repository of IF Department.

- _ To create and to query a relational database.
- _ To manage a Database Management System.
- _ To model distributed databases.

CONTENT

Part I: Relational Databases

- 1 - Introduction.
- 2 - Data Manipulation: using a declarative language [SQL]
- 3 - Programming: PL-SQL language
- 4 - Optimization [indexes, execution plan, cost model]
- 5 - Transactions [atomicity, consistency, isolation, durability].
- 6 - Database administration
- 7 - distributed databases: motivations
- 8 - distributed databases: conception
- 9 - data fragmentation
- 10 - data replication
- 11 - query processing in distributed databases
- 12 - transaction in distributed databases
- 13 - Conclusion, broader context

BIBLIOGRAPHY

Suzanne Dietrich, « An advanced course in database systems : beyond relational databases », Pearson/Prentice Hall, 2005.

Raghu Ramakrishnan, « Database management systems », McGraw-Hill, 2003.

S. Abitboul, R. Hull, V. Vianu « Foundations of Databases », Addison-Wesley 1995

Serge Abiteboul, Ioana Manolescu, Philippe Rigaux, Marie-Christine Rousset, Pierre Senellart « Web Data Management » Cambridge University Press 2012

Karl Aberer: Peer-to-Peer Data Management Morgan & Claypool Publishers 2011

Tamer Ozsu, Patrick Valduriez, «Principles of. Distributed Database Systems» 3rd edition, Springer, 2011

Netographie

S. Abitboul, R. Hull, V. Vianu « Foundations of Databases », Addison-Wesley <http://webdam.inria.fr/Alice/>

Le site du livre de R. Ramakrishnan et J. Gehrke <http://pages.cs.wisc.edu/~dbbook/>

PRE-REQUISITE

Module «Fondamentaux de la modélisation de données».

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